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Simulation of the inherent turbulence and wake interaction inside an infinitely long row of wind turbines

Søren Juhl Andersen*, Jens Nørkær Sørensen and Robert Mikkelsen

Department of Wind Energy, Technical University of Denmark, Lyngby, Denmark

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The turbulence in the interior of a wind farm is simulated using large eddy simulation and the actuator line technique implemented in the Navier–Stokes equations. The simulations are carried out for an infinitely long row of turbines simulated by applying cyclic boundary conditions at the inlet and outlet. The simulations investigate the turbulence inherent to the wind turbines as no ambient turbulence or shear is added to this idealised case. The simulated data give insight into the performance of the wind turbines operating in the wake of others as well as details on key turbulent quantities. One of the key features of wakes behind wind turbines is the dynamic wake meandering, which is shown to be related to the wind turbine spacing and the vortex shedding from the turbine as a bluff body. The flow is analysed and reconstructed by applying proper orthogonal decomposition.

Keywords: large eddy simulation; wind turbine wakes; actuator line; dynamic meandering; proper orthogonal decomposition

1. Introduction

In large wind farms, both onshore and offshore, the wind turbines in the interior will operate in the wake of the upstream turbines irrespective of the wind direction. A wind turbine operating in the wake of one or more wind turbines yields less power and is subject to greater loadings from the induced turbulence, leading to increased fatigue. The increased turbulence originates from the vorticity created by the rotor loading on the passing rotor blades. Understanding this highly dynamic wake interaction is very important for wind farm optimisation.

Several analytical wake models exist, e.g. Jensen [1] and Frandsen et al. [2]. Generally, these models are based on simple single wake calculations. Different assumptions are used to superpose merging wakes to describe the overall wake interaction inside wind farms. These methods are based on steady-state considerations and are developed with the aim of predicting mean quantities, e.g. the mean velocity deficit in the wake. Therefore, the models exclude the details of the dynamic wake interaction and turbulence properties needed to assess the turbine performance and loadings. Frandsen [3] proposed a model for fatigue loading on rotors using the effective turbulence intensity as the only governing parameter, arguing that the various changes in turbulence properties usually correlate with the standard deviation of wind speed fluctuations. Here the effective turbulence intensity is taken as a design variable, and not as a physical quantity. The model has shown improved accuracy compared to the existing engineering models, albeit still occasionally predicting

*Corresponding author. Email: sjan@dtu.dk

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large discrepancies compared to measurements. The analytical models have also been combined with models describing the so-called meandering or large-scale motion of wake. The dynamic wake meandering model assumes that the large-scale motions of the wake are directly caused by the large scales in the atmospheric turbulence, see Larsen et al. [4,5].

Measuring the wake interaction is not a trivial task and measurements contain large uncertainties due to the complex and everchanging inflow conditions, and hence the validation of the simpler engineering models is complicated. Therefore, numercial simulations are employed, where all relevant properties are described.

The wake interaction in wind farms has previously been studied using computational fluid dynamics and large eddy simulation (LES) employing the actuator disc and actuator line (AL) methods. These methods couple the full Navier–Stokes equations with body forces to include the effect of the wind turbines on the flow without having to fully resolve the boundary layers on the rotor blades. The Navier–Stokes-based actuator disc method was introduced by Sørensen and Myken [6] to simulate axisymmetric flows around wind turbines. An extension to full three-dimensional (3D) flows was later introduced as the AL technique developed by Sørensen and Shen [7]. These methods have been used extensively for wake studies on single and multiple wind turbines in wind farms, e.g. Troldborg [8] and Ivanell [9]. Recently, several studies have investigated large wind farms within the atmospheric boundary layer, e.g. Lu and Porté-Agel [10] and Calaf et al. [11]. The simulations have provided valuable insights into wake dynamics, but run times are long and thus unsuitable for practical use, e.g. to optimise wind farm layout, which requires numerous runs.

The trade-off between capturing the complex dynamics of the resolved flow and computational speed is still a pending issue when applying wake models to wind farm optimisation. The present work includes the analysis of numerical simulations of the dynamic and turbulent wakes deep inside an idealised wind farms with the aim of laying the foundation for more complex and physically correct wake models. Therefore, the present study excludes the effect of shear in the atmospheric boundary layer, thermal effects and gravitational effects in order to isolate and investigate the individual effects leading to the complex wake dynamics inherent to wind turbines, so that it will be easier to identify the various effects for more complex flow simulations. The key components of the highly complex flow will be identified by applying the proper orthogonal decomposition (POD), which will eventually form the basis for constructing a new reduced order wake model. The aim of such a reduced order model will be to settle the trade-off between including the dynamics and the high computational costs. Furthermore, this idealised case builds on the tradition of the simpler analytical models, such as that by Jensen [1].

2. Methods

2.1. *Numerical simulation*

2.1.1. *Flow solver*

The numerical simulations are performed using the 3D flow solver EllipSys3D, developed as a collaboration between Technical University of Denmark (DTU) (Michelsen [12]) and Risø (Sørensen [13]). This code solves the discretised incompressible Navier–Stokes equations in general curvilinear coordinates using a block structured finite volume approach. EllipSys3D is formulated in primitive variables (pressure–velocity) in a collocated grid arrangement. In the present work, the pressure correction equation is solved using the PISO algorithm and pressure decoupling is avoided using the Rhie/Chow interpolation technique. The

convective terms are discretised using a hybrid scheme combining the third-order accurate QUICK scheme and the fourth-order CDS scheme. This technique was employed as a compromise between avoiding unphysical numerical wiggles, occurring when using a pure fourth-order scheme, and limiting numerical diffusion due to the upwind biasing nature of the QUICK scheme.

The influence of the wind turbine is simulated using the AL technique. The AL technique applies body forces distributed along rotating lines representing the blades of the wind turbine, see Section 2.1.2 and Sørensen and Shen [7] for de- tails. The flow field is thus determined by solving the filtered (denoted by $\bar{V}$) 3D incompressible Navier–Stokes equations,

$$\frac{\partial \bar{\boldsymbol{V}}}{\partial t} + \bar{\boldsymbol{V}} \cdot \nabla \bar{\boldsymbol{V}} = -\frac{1}{\rho}\nabla \bar{p} + \nabla[(\nu + \nu_{SGS})\nabla \bar{\boldsymbol{V}}] + \frac{1}{\rho}\boldsymbol{f} + \frac{1}{\rho}\boldsymbol{f}_{mf}. \tag{1}$$

$$\nabla \bar{\boldsymbol{V}} = 0, \tag{2}$$

using LES (see Section 2.1.3) and applying the body forces, $\boldsymbol{f}$ and $\boldsymbol{f}_{mf}$, described in Section 2.1.2 and 2.1.4, respectively. ρ denotes density, p denotes pressure and ν denotes eddy viscosity. The AL method was implemented in EllipSys3D by Mikkelsen [14].

The advantage of representing the individual blades by line-distributed loadings is that much fewer grid points are needed to capture the influence of the blades, as compared to simulating the actual geometry of the blades. Therefore, the AL model allows for a detailed study of the dynamics of the different wake structures, such as the tip and root vortices, using a reasonably low number of grid points. Furthermore, the model benefits from being applicable with simple structured grids and therefore issues connected to grid generation do not occur. The drawback of the method, on the other hand, is that it relies on the quality of tabulated airfoil data.

2.1.2. *Turbine modelling*

The actual body forces used in the AL method are calculated using Flex5, a full aeroelastic code for calculating deflections and loads on wind turbines, see e.g. Øye [15]. The aerodynamic loads are calculated using a blade element approach including the dynamics of the induced velocities of the wake. The dynamics of the local inflow conditions are important to get realistic predictions of the response of changing the pitch angle, which forms the basis for the design of the control system for a pitch-regulated wind turbine.

Figure 1 shows the local velocities and forces acting on a cross-sectional airfoil segment. The local velocity triangle relative to the rotating airfoil segment is determined from the velocity triangle, where V_x and V_θ are the velocities in the axial and tangential directions, respectively, and the relative velocity is given as

$$V_{\text{rel}} = \sqrt{V_x^2 + (\Omega r + V_\theta)^2}. \tag{3}$$

The flow angle is given as $\phi = \tan^{-1}(\frac{V_x}{\Omega r + V_\theta})$, which gives a local angle of attack as $\alpha = \phi - \gamma$, with γ denoting the local pitch angle and Ω denoting the rotor speed. Lift and drag forces are found as

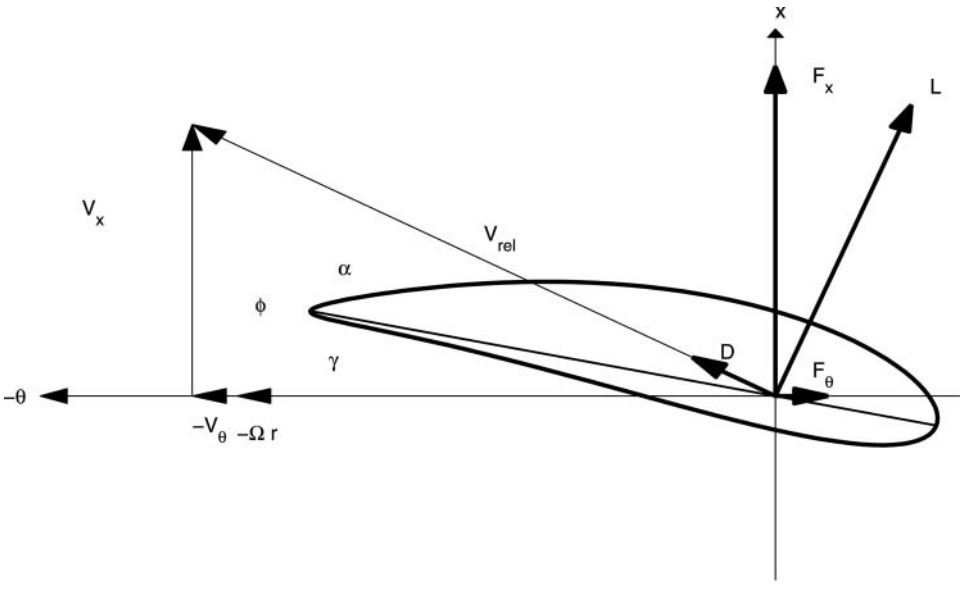


Figure 1. Velocity triangle showing velocities and force vectors used in the actuator line method.

$$\boldsymbol{f}_{2\mathrm{D}} = (\boldsymbol{L}, \boldsymbol{D}) = \frac{1}{2}\rho V_{\mathrm{rel}} c B (C_L \boldsymbol{e}_L, C_D \boldsymbol{e}_D), \tag{4}$$

where $C_L(\alpha, Re)$ and $C_D(\alpha, Re)$ are lift and drag coefficients as function of angle of attack (α) and Reynolds number (Re), given as tabulated data. The Reynolds number is based on chord length and relative velocity. e_L and e_D are unit vectors in lift and drag directions, respectively. Projection gives the axial and tangential forces, F_x and F_θ. The body forces are smeared across a few cells to avoid singularities. The smearing is Gaussian distributed by taking the convolution of the local load $\boldsymbol{f}_{2\mathrm{D}}$, and a regularisation kernel, η_ϵ, as follows:

$$\boldsymbol{f}(\boldsymbol{x}) = \sum_{i=1}^{B=3} \int_0^R \boldsymbol{f}_{2\mathrm{D}}(r) \eta_\epsilon(\|\boldsymbol{x} - r\boldsymbol{e}_i\|) dr. \quad \eta_\epsilon = \frac{1}{\epsilon^3 \pi^{3-2} \exp[-(r/\epsilon)^2]}, \tag{5}$$

where B is the number of blades, η_ϵ is a parameter controlling the regularisation and $r = \|\boldsymbol{x} - r\boldsymbol{e}_i\|$, where $\|\ \|$ is the Euclidean norm, i.e. the distance between the grid point and the force points on the $i'th$ AL denoted by the unit vector $\boldsymbol{e}_i$.

The airfoil data corresponding to the NM80 turbine is used in the present work, see Madsen et al. [16] for details. The two-dimensional (2D) airfoil data are corrected to account for 3D effects. The NM80 is a three bladed wind turbine with a radius $R = 40$ m and rated to 2.75 MW at $U_{\mathrm{hub}} = 14$ m/s, which is a scaled version of the original NM80 rated to 2.00 MW.

A controller is implemented for the turbine, which is a combination of a variable speed P-controller for the generator for wind speeds below rated ($U_{\mathrm{hub}} < 14$ m/s) and a PI-pitch angle controller for higher wind speeds. The variable speed controller sets the power set point based on the low-pass filtered generator speed, which basically means that

the power production is related to wind speed as $P \propto U^3$. The PI-pitch angle controller keeps a constant power production by changing the global blade pitch. Details of how a similar controller operates can be found in Hansen et al. [17]. The implemented controller essentially means that the rotor is not constantly loaded, and as such it operates like a real turbine.

2.1.3. *Large eddy simulation*

The scales resolved by the grid are simulated directly by the Navier–Stokes equations, whereas scales below the grid scale is modelled using LES. The sub-grid scale (SGS) viscosity is modelled using the vorticity-based mixed scale model by Ta Phuoc et al. [18]. In this model the SGS viscosity is given by

$$\nu_{SGS}(\beta)(\boldsymbol{x}, f) = \rho C_m |\nabla \times \bar{\boldsymbol{V}}(\boldsymbol{x}, t)|^{\beta} \left(q_c^2\right)^{\frac{1-\beta}{2}} (\boldsymbol{x}, t) \bar{\Delta}^{1+\beta}, \tag{6}$$

where ρ is the density of air, $\bar{\boldsymbol{V}}$ is the filtered velocity, $\bar{\Delta}$ is the filter cut-off length, which is set equal to $\Delta \mathrm{Vol}^{1/3}$, where ΔVol is the volume of a given cell. C_m and β are constants set to 0.01 and 0.5, respectively.

The kinetic energy q_c^2 is evaluated as

$$q_c^2(\boldsymbol{x}, t) = \frac{1}{2} \left(\bar{\boldsymbol{V}}(\boldsymbol{x}, t)\right)' \left(\bar{\boldsymbol{V}}(\boldsymbol{x}, t)\right)', \tag{7}$$

where $(\bar{\boldsymbol{V}})' = \bar{\boldsymbol{V}} - \tilde{\bar{\boldsymbol{V}}}$ represents the high frequency part of the resolved velocity field. This field is defined using a second filter ($\tilde{\Delta} = 2\Delta$), referred to as the test filter denoted by $\tilde{}$. The mixed scale model has been chosen because of its simplicity and because it accounts for the dissipation of energy at the same time as it ensures the interaction between the smallest resolved scales and the largest unresolved scales (for a general overview of models, see Sagaut [19]).

2.1.4. *Domain set-up*

The wake interaction is examined in an idealised case of an infinitely long row of turbines with initial uniform inflow of $U_0 = 15$ m/s, which is above rated speed for the wind turbine. Uniform inflow means that no atmospheric boundary layer and no ambient turbulence is added, making it possible to study the turbulence inherent to the wind turbines. Furthermore, the situation is idealised by excluding ground, gravitational and temperature effects on the flow. The infinitely long row of turbines comes from applying cyclic boundary conditions in the flow direction, i.e. coupling the inflow condition to the outflow.

The cyclic domain consists of a 162-block structure in a Cartesian grid with a total of approximately 18×10^6 grid points. The domain is $36R \times 20R \times 20R$ in X, Y, Z, respectively, where X is the streamwise, Y the lateral and Z the vertical direction. The size of the domain is chosen to avoid numerical blockage by adhering to a blockage ratio of 0.8%, which is less than 3% as recommended by Baetke et al. [20], and to limit if possible the introduction of additional frequencies due to the domain, i.e. ensuring that the domain for the flow direction is dominant. Three turbines are placed in the domain, with a spatial distance between the turbines of $12R$. The reason for using three turbines instead of one is to avoid constraining the development of the turbulence due to the finite size of the domain.

Introducing three turbines ensures that the turbulent length scales of size up to at least $36R = 1440$ m may appear in the computations. The turnaround time is defined as the time it takes for the free stream wind to pass through the domain. With a rotor radius of $R = 40$ m and three turbines with a distance of $12R$ between the turbines, the turnaround time is $T_T = \frac{3\times12\times40\,\mathrm{m}}{15\,\mathrm{m/s}} \approx 96$s. Far field Dirichlet boundary conditions ($U =$ constant) are applied on the transverse and vertical boundaries, i.e. making it a single row of wind turbines, not an infinite wind farm in the transverse direction. The grid is equidistant in the streamwise direction, which is necessary to maintain and model the turbulence correctly as it is transferred using the cyclic boundary conditions. The grid is non-equidistant in the transverse and vertical directions with a higher resolution in an equidistant region for $(y, z) = [\pm 3R]$, where the rotors are located in $(x, y, z) = (4R, 0R, 0R), (16R, 0R, 0R), (28R, 0R, 0R)$, see Figure 2, where the rotor is indicated by a circle and Figure 5 for a 2D top view. Each rotor blade is resolved by 20 grid points.

The nature of the simulations with cyclic boundary conditions and long run times would usually lead to a loss in mass flux as the turbine extracts energy from the flow. However, the mass flux is continuously corrected to account for the energy extracted by the turbines by applying small body forces ($\boldsymbol{f}_{mf}$) everywhere in the flow, essentially corresponding to a pressure gradient. This correction is calculated using a Proportional-Integral-Derivative (PID) controller to correct for the error. The error (ξ_j) for a given time step j is defined as

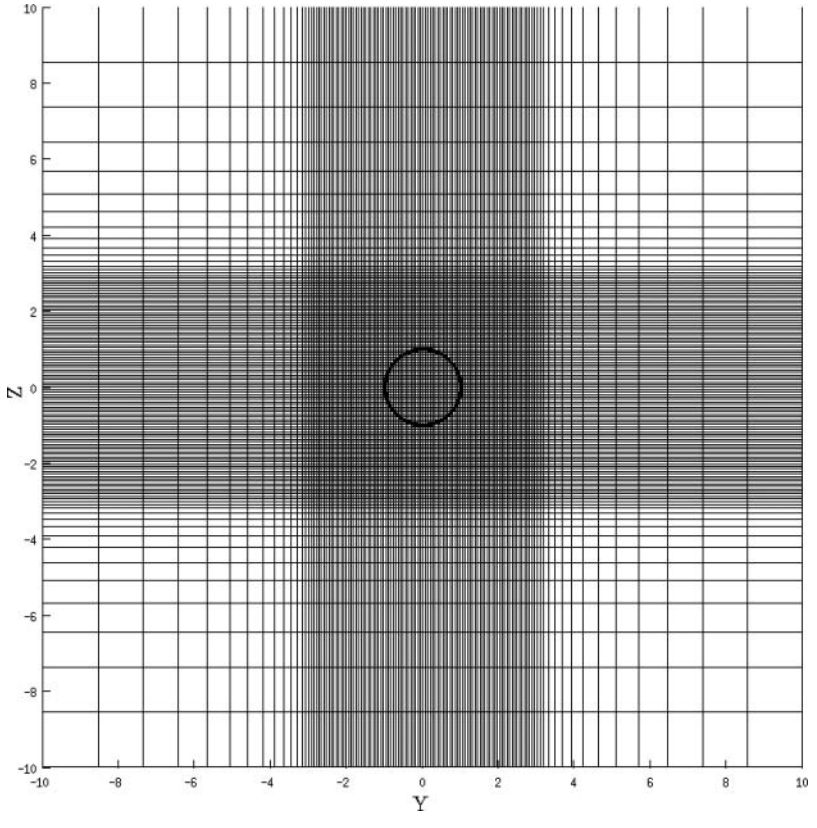


Figure 2. Cross-sectional mesh.

the difference between the initial mass flux (ϕ_0) and the mass flux at time j, (ϕ_j),

$$\xi_j = \phi_0 - \phi_j = \int_S \rho \boldsymbol{u}_0 d\boldsymbol{S} - \int_S \rho \boldsymbol{u}_j d\boldsymbol{S}, \tag{8}$$

where $\boldsymbol{S}$ is the surface. The PID controller integrates over the previous 20 time steps, so the correcting body force at time j is given as

$$\boldsymbol{f}_{mf,j} = K_p \xi_j + K_i \int_{t=j-19}^{t=j} \xi dt + K_d \frac{d\xi_j}{dt}. \tag{9}$$

The standard coefficients K_p, K_i and K_d of a PID controller has been set to 2, 1 and 1, respectively. This ensures that the mass flux is kept constant during the simulation time.

2.2. *Analysis methodology*

The analysis of the inherent turbulence focuses on data extracted immediately upstream the first turbine ($1R$), i.e. the incoming flow as seen by the turbine (see Figure 5 where the three turbines and the extracted plane is shown). Slices of data are extracted every 0.107 s. The wake centre is identified using a mass centre analogy on the wake deficit ($1 - \frac{U}{U_0}$) within the equidistant region,

$$y_C = \frac{\int y(1 - \frac{U}{U_0}) dA}{\int (1 - \frac{U}{U_0}) dA}, \qquad z_C = \frac{\int z(1 - \frac{U}{U_0}) dA}{\int (1 - \frac{U}{U_0}) dA}, \tag{10}$$

where dA is the cross-sectional area. This describes the large-scale motion of the wake. The large-scale motion of the wake is sometimes referred to as the wake meandering, see Larsen et al. [5]. Standard definitions (see Pope [21]) are used to calculate Reynolds stresses, turbulent kinetic energy (TKE) and TKE production although density, ρ, is excluded. The Reynolds stresses are calculated as

$$\tau_{ij} \equiv \langle u_i' u_j' \rangle, \tag{11}$$

where $\langle\rangle$ denotes the time average, while TKE is half the trace of the Reynolds stress tensor,

$$k \equiv \frac{1}{2} \langle u_i' u_i' \rangle, \tag{12}$$

The production of TKE is calculated as

$$\mathcal{P} \equiv -\langle u_i' u_i' \rangle \frac{\partial \langle u_i \rangle}{\partial x_j}. \tag{13}$$

In the present case, all velocities have been normalised by U_0 prior to calculating the turbulent quantities, hence unitless.

2.2.1. *Proper orthogonal decomposition*

POD is a statistical method yielding an optimal linear subspace, optimal in terms of the variance of the energy content, i.e. it ensures that the sorted modes contain the most energetic states. POD was first introduced as an analysis tool for turbulence studies by Lumley [22], and POD is a proven method for examining coherent structures for stationary random processes. The decomposition starting point is a Hermitian symmetric two-point velocity (fluctuations) correlation tensor:

$$R_{i,j}(\boldsymbol{x}, \boldsymbol{x}', t, t') = \overline{u_i(\boldsymbol{x}, t), u_j^*(\boldsymbol{x}', t')} \tag{14}$$

and subsequently the integral eigenvalue equation of Fredholm type is solved,

$$\int R_{i,j}(\boldsymbol{x}, \boldsymbol{x}', t, t')\mathbf{g}_j(\boldsymbol{x}', t') = \lambda \mathbf{g}_j(\boldsymbol{x}', t'). \tag{15}$$

The solution yields an optimal subspace given as a set of real eigenvalues:

$$\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \cdots \geq \lambda_N \mathrm{a} \tag{16}$$

and orthogonal eigenfunctions. For additional details, reference is made to the original work by Sirovich [23], Aubry et al. [24] and the general overview given by Berkooz et al. [25]. A brief overview of the practical implementation of POD is given in the following, adapted from Jørgensen et al. [26]. POD is applied to all three velocity components within the equidistant region, as there should otherwise be added a spatial weighting matrix. The normalised velocity components ($\boldsymbol{u} = (u, v, w)$) of each of the N extracted slices are organised in vectors $\boldsymbol{u_i}$, which constitute a matrix defined as

$$\boldsymbol{U} = [\boldsymbol{u_1'} \ldots \boldsymbol{u_N'}]. \tag{17}$$

It is common to subtract the mean value $\boldsymbol{u_0}$, hence reducing the number of dimensions by one. Therefore, $\boldsymbol{u_j'}$ is defined as

$$\boldsymbol{u_j'} = \boldsymbol{u_j} - \frac{1}{N}\sum_{i=1}^{N} \boldsymbol{u_i} = \boldsymbol{u_j} - \boldsymbol{u_0}. \tag{18}$$

The $N \times N$ auto-covariance matrix is then given as

$$\boldsymbol{R} = \boldsymbol{U^T U}. \tag{19}$$

Defining an eigenvalue matrix with $\lambda_1 \geq \lambda_2 \geq \cdots \lambda_{N-1} > 0$:

$$\boldsymbol{\Lambda} = \begin{pmatrix} \lambda_1 & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & \lambda_{N-1} \end{pmatrix}$$

and matrix of orthonormal eigenvectors,

$$\boldsymbol{G} = [\boldsymbol{g_1} \ldots \boldsymbol{g_{N-1}}] \tag{20}$$

yields a eigenvalue problem of the form,

$$\boldsymbol{RG} = \boldsymbol{G\Lambda}. \tag{21}$$

The basis vectors or POD modes can be written as

$$\phi_k = \frac{\boldsymbol{U g_j}}{\left\| \boldsymbol{U g_j} \right\|}. \tag{22}$$

These are spatial modes, which contain information on the turbulent structures. It is important to note that these individual structures or POD modes might not be physical structures in the sense that they are visible to an observer, but merely a result of the mathematical operations lumping the energy content into an optimal set of modes.

Hence, the original vectors $\boldsymbol{u'_j}$ containing the velocity components of slice $\boldsymbol{j}$ can be reconstructed from

$$\boldsymbol{u'_j} = \sum_{k=1}^{N-1} \phi_k a_{kj}, \tag{23}$$

where a_{kj} are the elements of $\boldsymbol{A} = \boldsymbol{\Phi^T U}$. $\boldsymbol{A}$ contains the temporal development of the spatial POD modes. A future development of a reduced order model will be based on a truncated reconstruction, where only the most important modes are included.

3. Results

3.1. *Turbine performance history*

The turbulence inherent to the wind turbines is related to the performance of the wind turbine. Figure 3 shows the streamwise velocity (and the filtered mean) as function of time at a hub height of $1R$ upstream of the third rotor. It is worth noticing the strong initial decrease from unity to $\frac{U_{\text{hub}}}{U_0} \approx 0.2$ in the early stages, i.e. when the flow goes through the first wind turbines. The incoming wind velocity recovers from the initial large decrease and is almost constant (only very small increase over time) at $\frac{U_{\text{hub}}}{U_0} \approx 0.7$ from $t = 1100$ s and onwards. The mean velocity being almost constant is a good sign for the validity of analytical models, such as the models by Jensen [1] and Frandsen et al. [2]. Figure 4 shows three quantities describing the turbine performance: wind speed at hub height, tip speed and the power production for $t \in [2100s, 2700s]$. The pitch is not shown as it generally attains a constant value of 0.1° as soon as the turbines are deep within the wind farm, i.e. as the wind turbine operates below rated. The velocity at hub height fluctuates quite significantly, which also gives rise to fluctuations in the power production. Despite the large fluctuations in velocity, the controller ensures that the fluctuations in the power production is less violent in order to decrease the fatigue loads on the generator and turbine. Standard deviation on the normalised velocity is $\sigma(\frac{U_{\text{hub}}}{U_0}) = 0.11$ and normalised power production is $\sigma(\frac{P}{P_0}) = 0.13$ as compared to $\sigma((\frac{U_{\text{hub}}}{U_0})^3) = 0.17$ since the incoming wind is mostly below rated where $P \propto U^3$. The tip speed is rather constant around $\Omega \cdot R \approx 72$ m/s, where $\Omega \approx 1.80$ rad/s is the rotational speed. The highly complex flow in the wake clearly gives rise to a very complex behaviour of the wind turbine itself, which again feeds more complexity into the flow.

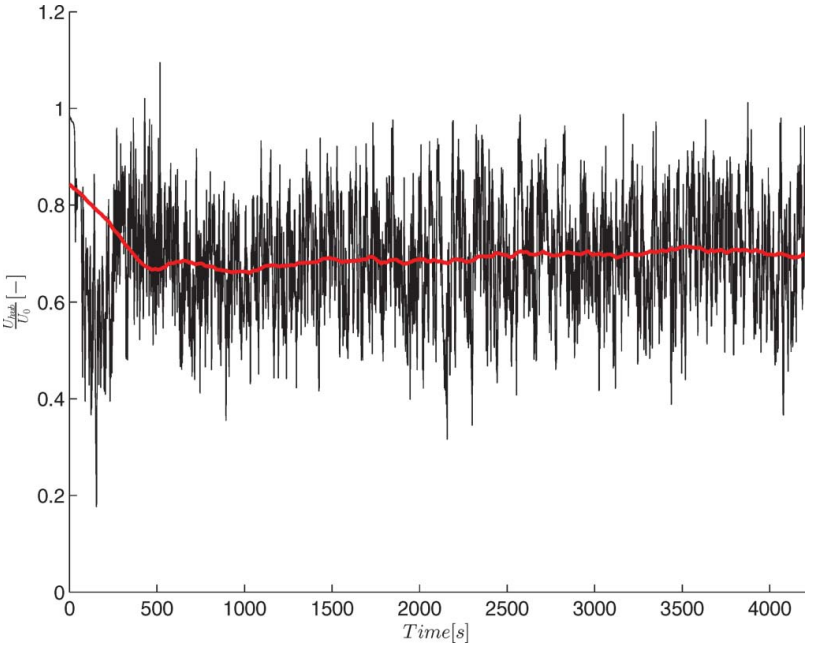


Figure 3. Streamwise velocity at hub height $1R$ upstream and filtered velocity.

3.2. *Turbulent quantities in the wake*

Figure 5 shows an isoplot of the instantaneous vorticity of the highly turbulent flow between the three turbines. It is particularly worth noting how the wake has moved upwards between the second and third turbines. Figure 5 also indicates the plane $1R$ upstream the first turbine, where the data for the following analysis have been extracted. Hence, the data correspond the incoming wind as seen by a wind turbine deep inside a wind farm and the wake $11R$ downstream the previous turbine.

Figure 3 indicates how the flow between the turbines approaches an equilibrium. Ideally, the flow should approach a random stationary process in order to apply POD. Therefore, the turbulent quantities should converge statistically. The flow is assumed to have reached its equilibrium after $t = 1100$ s, and the turbulent quantities are calculated for a range of different temporal windows for $\pm 3R$. The spatial root-mean-square (RMS) value of these quantities are used to show that the quantities converge to a constant level. Figure 6 shows the mean velocities and RMS values of the velocity fluctuations. Clearly, the streamwise velocity dominates the flow, with close to zero lateral and vertical components. The minor discrepancy from zero is related to the swirl in the wake due to the rotation of the blades. It is worth noting that the values of the mean streamwise velocity ratio here is significantly larger than 0.7 seen in Figure 3, because it is averaged for $\pm 3R$, i.e. it includes high velocity deficits, which are often outside the main wake, as described in the previous section. The streamwise fluctuations are approximately 25–30% larger than the inplane fluctuation. Generally, the mean velocities are approximately constant, while it requires larger windows (≈35 min) for the streamwise velocity fluctuations to reach a plateau with a level of approximately $\sqrt{\bar{u'^2}} = 0.103$.

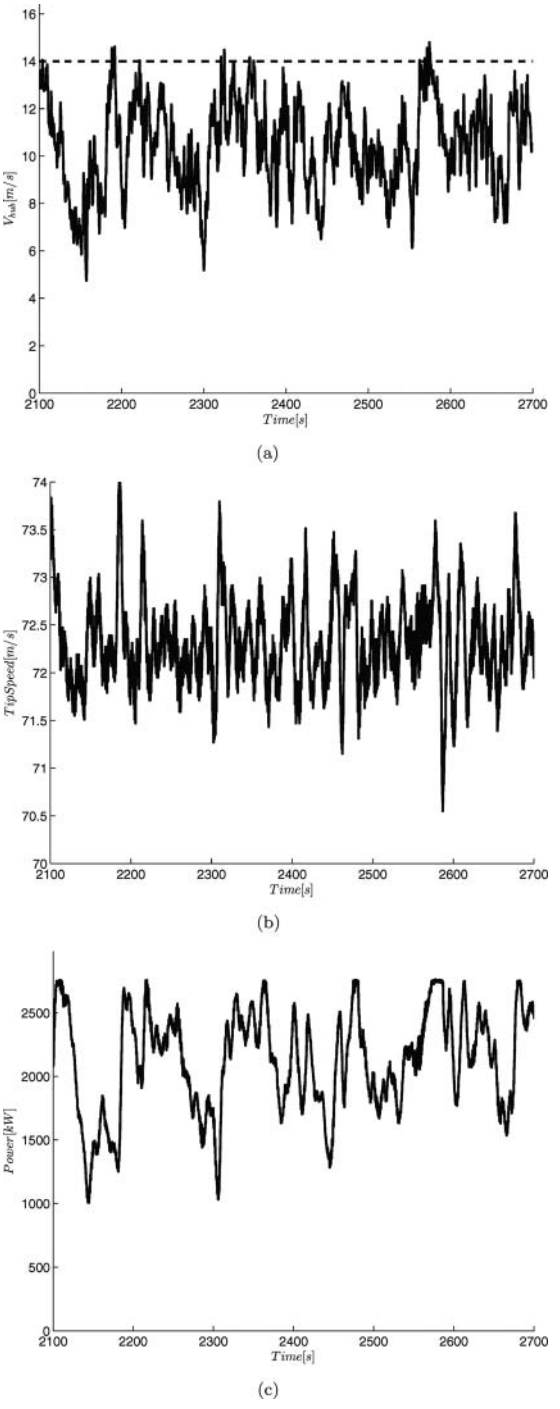


Figure 4. Wind speed at hub height, tip speed and power production in an infinitely long row of wind turbines for the period $[900s - 1500s]$. (a) Wind speed at hub height. Rated wind speed is indicated with the broken line. (b) Tip Speed. (c) Power production.

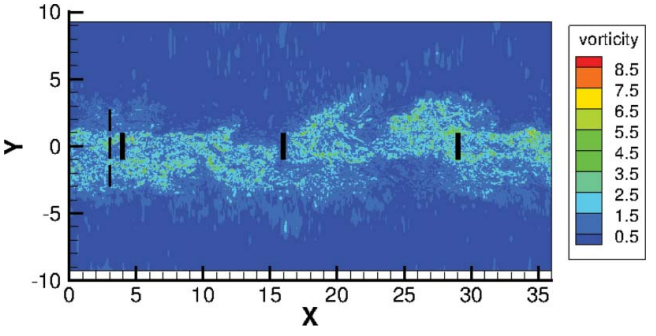
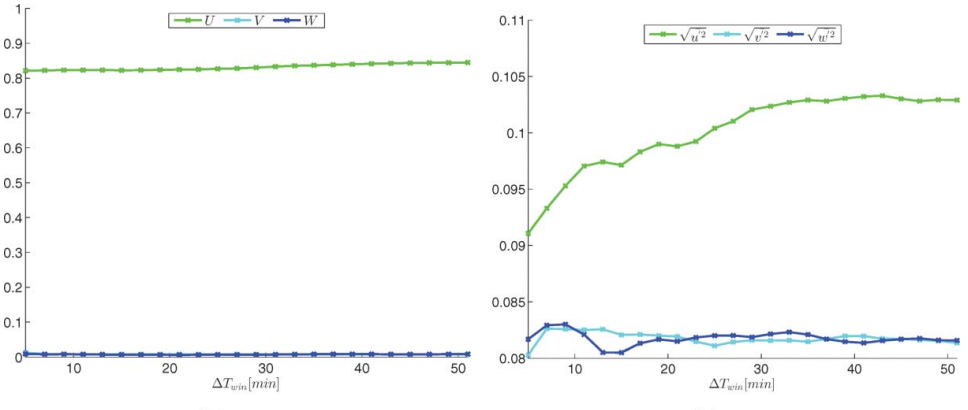


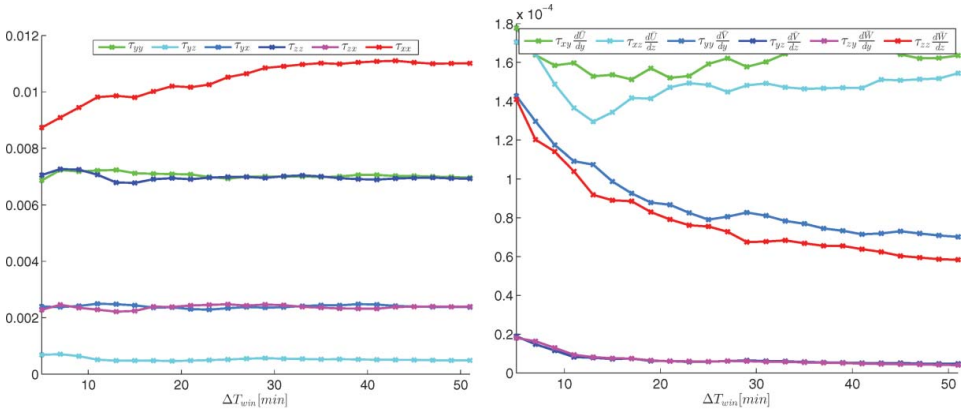
Figure 5. Isoplot of vorticity of the flow between the three turbines, which are marked by black lines. The broken line indicates the extracted slices used in the analysis. Inflow from the left.

Similarly, Figure 7 shows the convergence for the inplane Reynolds stresses and TKE production. The diagonal Reynolds stresses ($\tau_{i,j}, i = j$) dominates the flow. The Reynolds stresses clearly portrait a similar convergence as the mean velocities, as expected all the Reynolds stresses have also reached a constant level after ≈35 min. The TKE production is dominated by the components related to the streamwise velocity, which reaches an approximate constant level after ≈25 min. The terms related to the diagonal Reynolds stresses are still decreasing slightly after averaging over more than 50 min of real time. The slight decrease in TKE production is associated with a slight increase in mean streamwise velocity at hub height, i.e. the mean flow appears to regain some energy as indicated in Figure 3. The last two terms have also reached a constant and much smaller level. Since the terms stemming from the streamwise velocity ($\tau_{xy}\frac{dU}{dy}$ and $\tau_{xz}\frac{dU}{dz}$) dominate the TKE production, the slight decrease is only assumed to play a minor role in the overall statistical convergence and the flow is therefore assumed to have reached a random stationary process when averaging for longer than 35 min, since the dominant quantities have converged towards an acceptable convergence in terms of constant level. Reaching a perfectly symmetric and converged solution would require extremely long run times. Therefore, 35 min will in the following be used as the converged period.



(a) x_C. (b) y_C.

Figure 6. Spatial RMS values for the temporal convergence of the mean and RMS velocities for varying window lengths. (a) Mean velocities [−]. (b) RMS velocity fluctuations [−].



(a) x_C. (b) y_C.

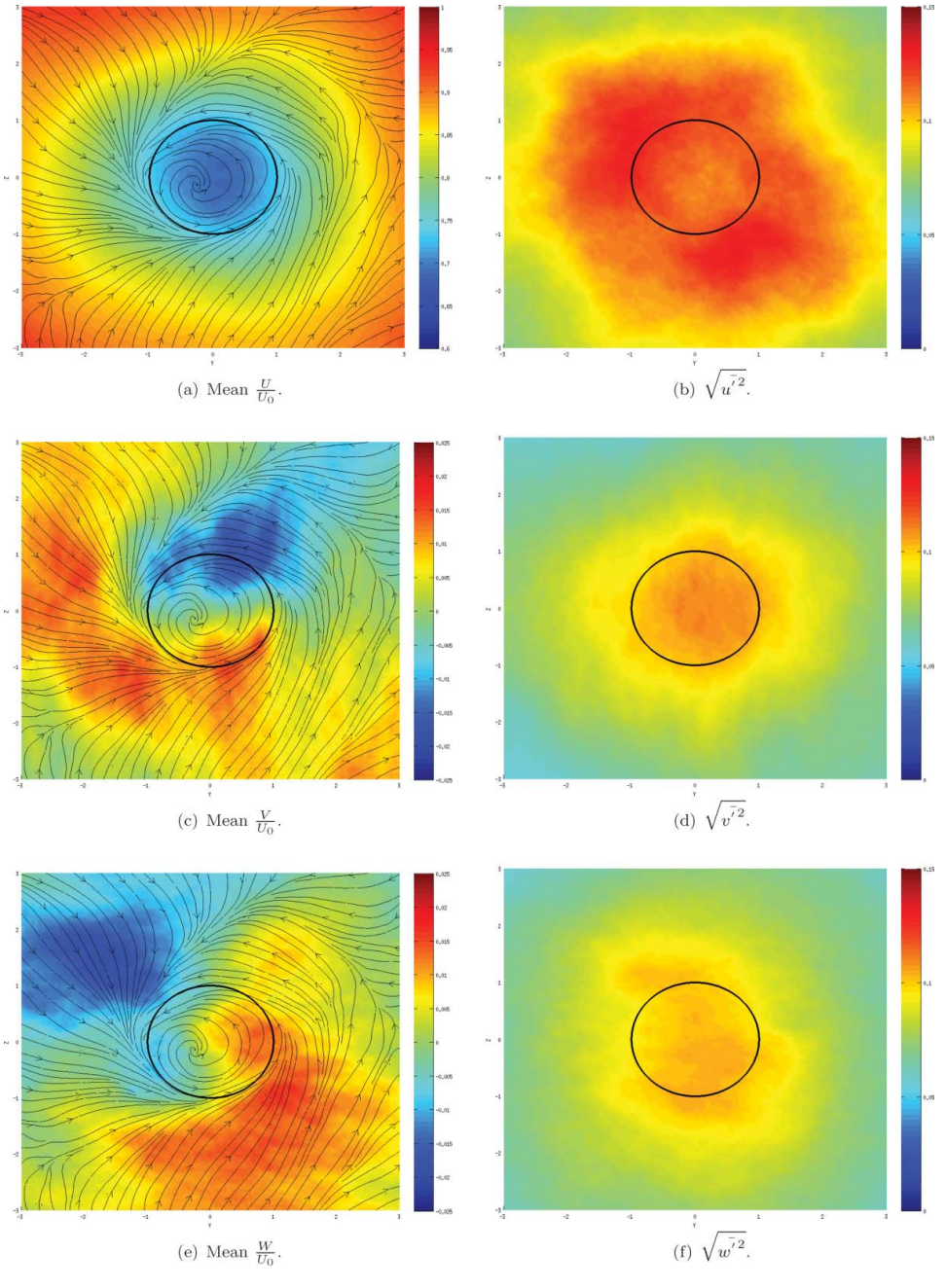
Figure 7. Spatial RMS values for the temporal convergence of the Reynolds stresses and TKE production for varying window lengths. (a) Reynolds stresses [−]. (b) TKE production [−].

Figure 8 shows the mean and RMS values of the fluctuations of the three velocity components. The flow is naturally dominated by the streamwise velocity with a clear symmetric appearance. The lateral and vertical mean velocities are less distinct and are not symmetric. The minor non-symmetries are attributed to the very long convergence time needed to obtain complete symmetry in the statistics. However, these components are only minor and they contain the swirl of the wake, which in average is counterclockwise as expected behind a wind turbine. The streamlines in the plane have been plotted on top of the mean velocities for clarity. The velocity fluctuations are of the same order of magnitude for all three components, with the highest level for the streamwise fluctuations. The streamwise fluctuations are spread over a larger area ($\pm 3R$), while the inplane fluctuations are concentrated within and around the rotor area of $\pm 2R$. It is worth noting, how the flow deep inside a wind farm no longer contains dominant ring structures usually seem in the near wake in AL simulations, because the data is extracted outside the near wake. The tip and root vortices are broken down by the inherent turbulence generated by previous turbines. Similar turbulent break down by atmospheric turbulence has previously been reported by Troldborg [8].

Figure 9 shows the Reynolds stresses. The diagonal Reynolds stresses, τ_{xx}, τ_{yy} and τ_{zz}, dominate the overall Reynolds stresses, and reveal similar patterns as those for the fluctuations in Figure 8. There is a clear symmetry in both the off- diagonal Reynolds stresses involving the streamwise fluctuations, i.e. τ_{yx} and τ_{zx}, which shows how momentum is transferred into the wake centre.

The symmetry in τ_{yz} and τ_{zx} are clearly due to the Cartesian nature of the current plots. The Reynolds stress (τ_{yx}) in the rotor plane is more evenly distributed, but still symmetric about the diagonal of the flow with positive Reynolds stress in quadrants 1 and 4 and negative in quadrants 2 and 3.

Figure 10 yields the total TKE production in the rotor plane, i.e. the sum of the terms depicted in Figure 7 and excluding the production terms including the streamwise derivatives, $\frac{d}{dx}$. Clearly, there are two major sink and two major source regions located around and immediately outside the rotor radius, i.e. underlining how momentum is being transferred into the wake.



(a) Mean $\frac{U}{U_0}$. (b) $\sqrt{\overline{u'^2}}$.

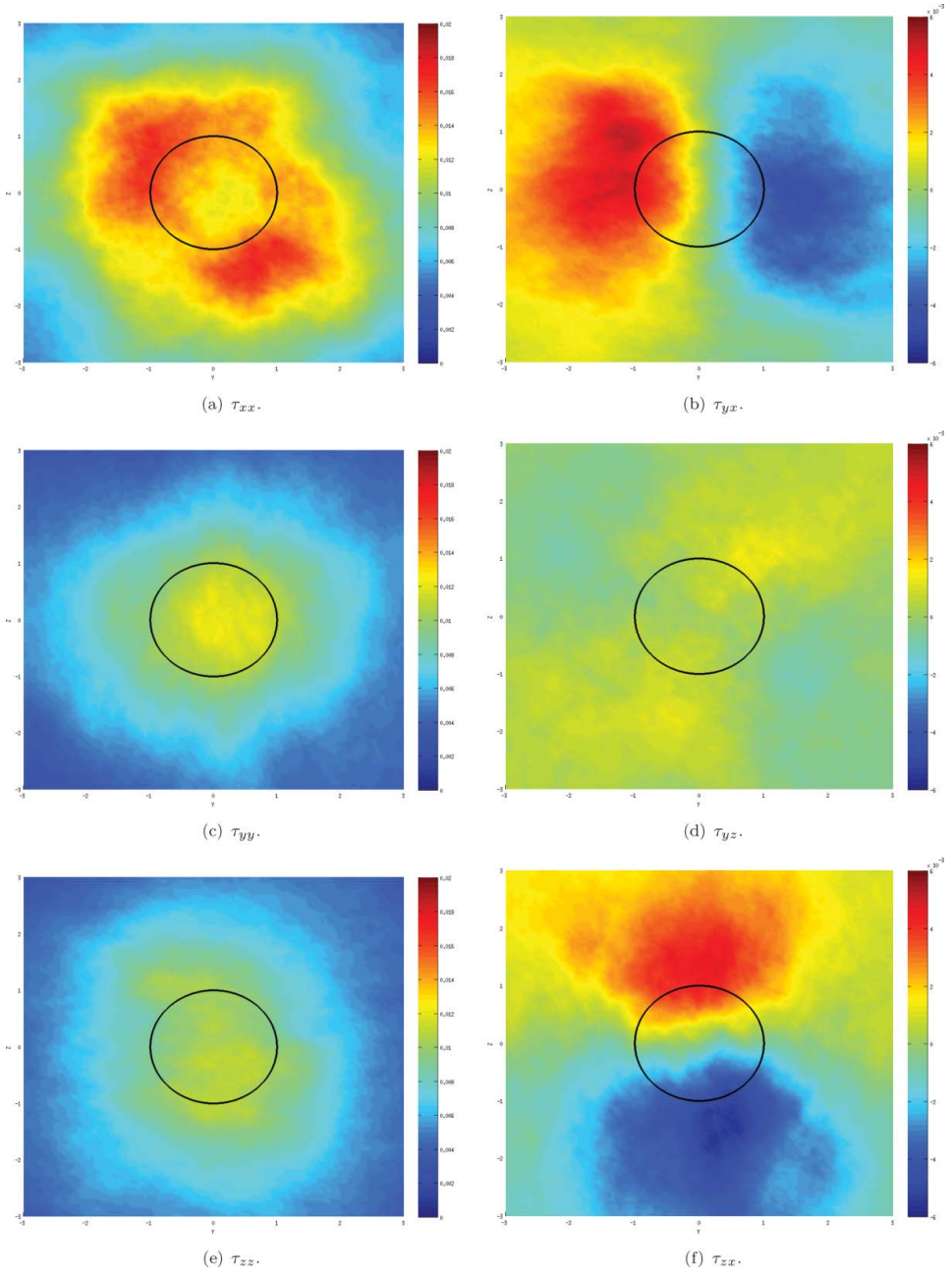
(c) Mean $\frac{V}{U_0}$. (d) $\sqrt{\overline{v'^2}}$.

(e) Mean $\frac{W}{U_0}$. (f) $\sqrt{\overline{w'^2}}$.

Figure 8. Mean velocities and RMS of the velocity fluctuations normalised by U_0. Note the difference in scales.

3.3. *Dynamic meandering of the wake center*

Figure 11 shows an example of an instantaneous slice of the streamwise velocity ($\frac{U}{U_0}$) extracted $1R$ upstream the first turbine. The dot indicates the hub of the oncoming wind turbine and the broken line indicates the rotor area. The cross marks the wake centre



(a) τ_{xx}. (b) τ_{yx}.

(c) τ_{yy}. (d) τ_{yz}.

(e) τ_{zz}. (f) τ_{zx}.

Figure 9. Reynolds stresses. Note that the diagonal Reynolds stresses has a different scale compared to the off-diagonal.

(yC, zC) along with a $2R$ circle marked with a full line. The flow is clearly highly complex with small and large scales, and the wake does not hit the following wind turbine straight on. There is also a significant wake deficit on the right side of the figure, which is related to the reminiscents of other wakes. It is no longer only one wake, but the combined effect of numerous wakes and the combined interaction. Furthermore, the complex flow clearly

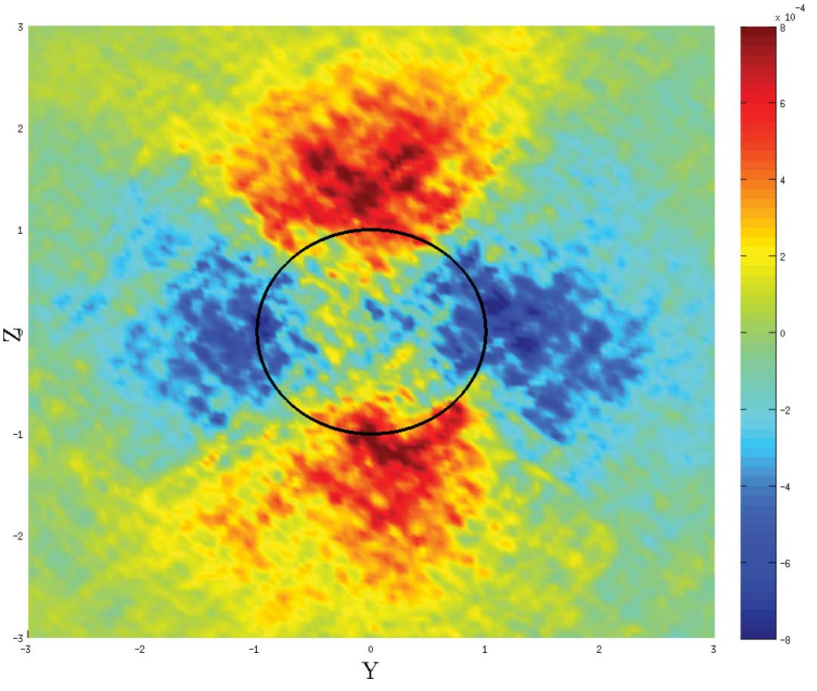


Figure 10. Total TKE production in the plane.

gives rise to differentiated loading on the turbine due to the larger wake deficit in the lower left side compared to the upper right corner of the rotor area, which again leads to the highly complex performance of the wind turbine described previously.

The motion of the wake centre coordinates (yC, zC) is shown in Figure 12. During the initial turnarounds, the amplitudes of the motion grow before reaching a peak level of approximately 1–1.5R with a few extreme movements as far as 2R from the hub centre. The standard deviations of the lateral and vertical directions are 0.63 and 0.65, respectively. This validates that there should be no preferred direction in the system, i.e. no stratification from gravity or temperature effects and same far field boundary conditions on vertical and lateral boundaries. Furthermore, the centre of the wake does not always rotate clockwise or counterclockwise, as the wake immediately behind a wind turbine, which rotates opposite the wind turbine. However, the wake rotation changes direction in a non-predictive manner. The trajectory of the wake centre is illustrated in Figure 13 for a brief period, where the movement of the wake changes from a clockwise to a counter-clockwise rotation. This surprising behaviour is underlined by the mean spanwise and vertical velocity patterns shown in Figure 8.

From Figure 12, it is evident that the wake centre moves with some dominant frequencies. To determine the frequencies, the time series ($t \in [1100s - 7200s]$) has been split into 7 overlapping windows of 35 minutes and a Hamming window is applied. Spectral analysis has been performed on the filtered signals for all periods and then averaged across the windows, see Figure 14. The spectral analysis reveals a distinct peak frequency around $f_{P,1} \approx 0.009$ Hz for both the horizontal and vertical motions, as well as additional peaks at $f_{P,2} \approx 0.017$ Hz and $f_{P,3} \approx 0.025$ Hz. The first two frequencies ($f_{P,1}$ and $f_{P,2}$) are assumed related to the turbine spacing, which gives rise to frequencies similar to those

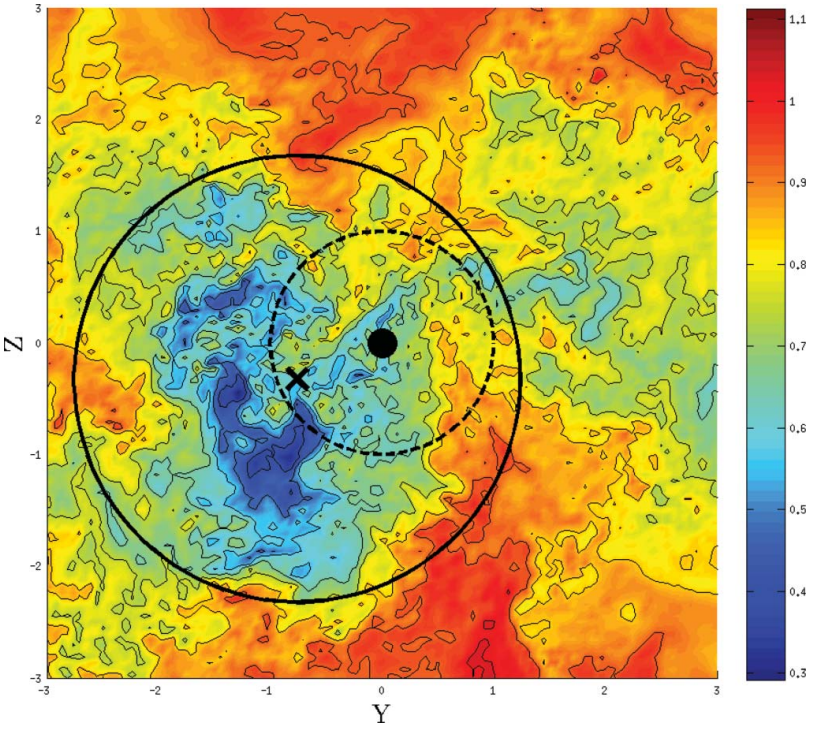
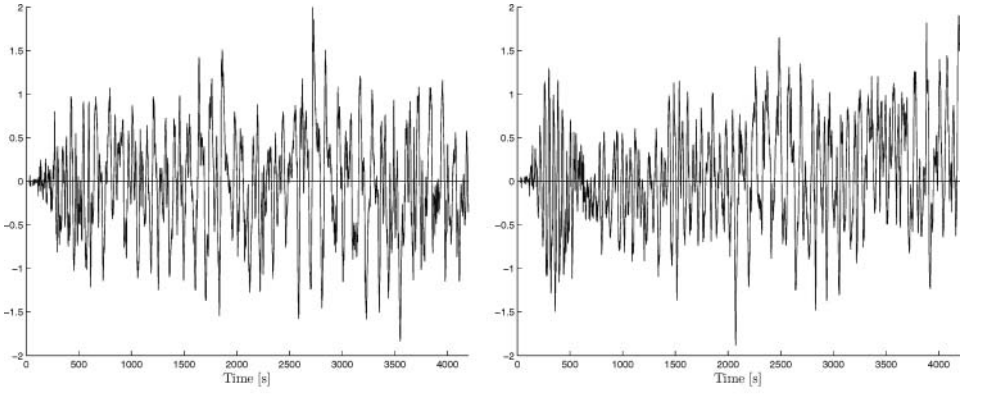


Figure 11. Instantaneous slice of $\frac{U}{U_0}$ contours at $1R$ upstream the first turbine. The filled circle and broken line depict the hub and rotor area of the following wind turbine $1R$ downstream. The cross shows the centre of mass and the full line $2R$ around (x_C, y_C) , i.e. the wake in the current definition.

reported by Lu and Porté-Agel [10] and related to the atmospheric turbulence. However, there is no atmospheric turbulence in the present case, but the large-scale turbulence generated will be related to the turbine spacing and of the same magnitude as atmospheric turbulence $O(500)$ m, see Frandsen [3]. The frequencies could potentially be related to the domain size, but it would need a much larger simulation to verify either one. The latter frequency corresponds to a Strouhal number of $St_3 = \frac{f_{P,3}2R}{0.7U_0} = 0.19$, corresponding to what is usually obtained for the vortex shedding behind a bluff body. Previously, Medici and Alfredsson [27] have shown how turbines shed structures as a bluff body with $St = 0.12$ for a tip-speed ratio above 4.5. Hence, we conclude that the large-scale motions are mainly governed by the turbine spacing and to a lesser degree the equivalent bluff body shedding.

The normalised TKE power spectral density at $1R$ upstream the first turbine for $t \in [1100s - 3200s]$ is averaged spatially within $\pm 3R$ and shown in Figure 15 together with the power spectral density of the velocity fluctuations. For comparison, the curves are derived and shown both in the fixed frame of reference and the moving frame of reference, i.e. subtracting the wake centre. The TKE power spectral density shows a clear low frequency peak at approximately $f = 0.008$ Hz and $f = 0.019$ Hz, i.e. almost identical to those found for the wake centre. The difference is that here is averaged over a larger spatial area. The same peaks are present in the moving frame of reference, which simply are shifted due to the shift in system of reference. The same is evident from the power spectral density of the velocity fluctuations. This shows that low-frequency dynamics are not only related to the large scales of the atmospheric turbulence as implicitly assumed in dynamic



(a) y_C. (b) z_C.

Figure 12. Wake centre motion as calculated using the centre of mass approach. (a) Horizontal displacement of the wake centre. (b) Vertical displacement of the wake centre.

wake meandering models. The large-scale motions are an integrated and inseparable part of the entire wake dynamics. A more sophisticated way of determining the wake centre may be proposed, although several techniques have been tried in the current study. The highly complex wake dynamics deep within a wind farm should not be modelled as a collective entity overlaid with the large-scale motions of the atmospheric turbulence. The present results shows that this assumption is only valid for the first few turbines, but does not hold in the interior of a wind farm. This is in part due to the instantaneous lack of axisymmetry, also indicated in Figure 11 and similar instantaneous flow field, which shows

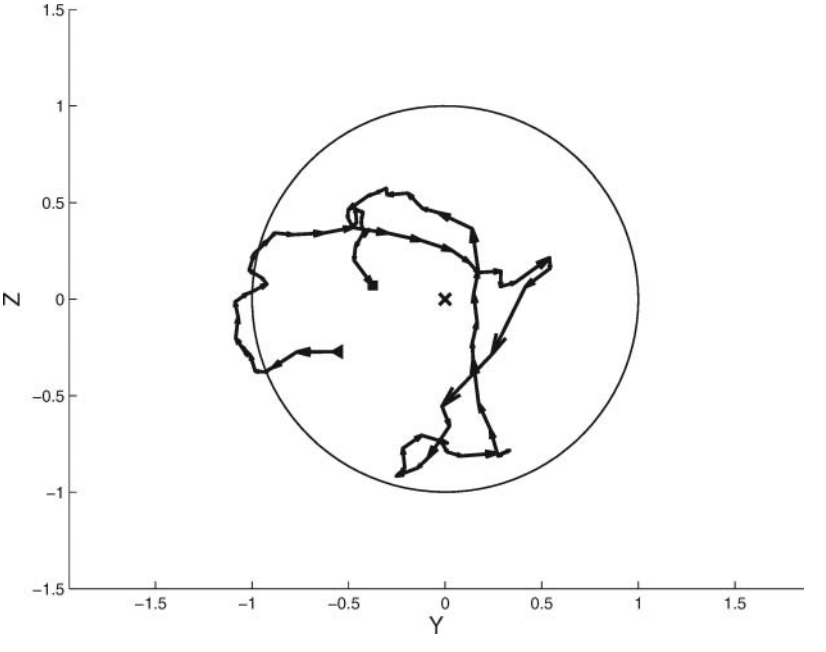
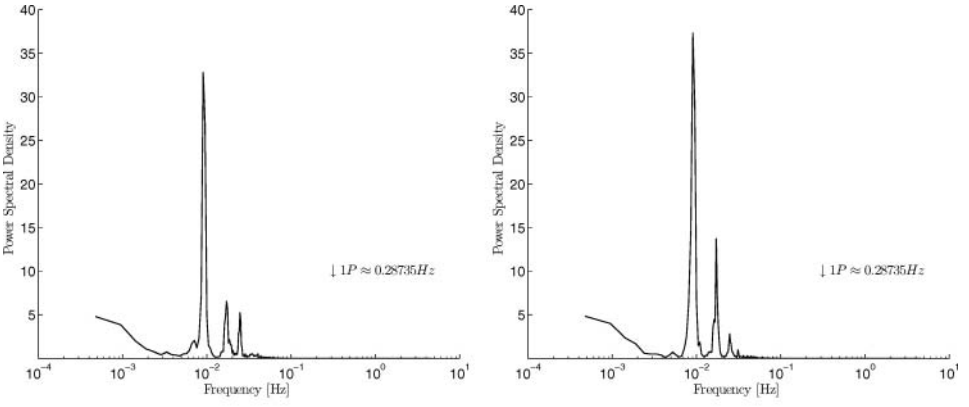


Figure 13. Trajectory of the wake centre from $t = 1050$ s to $t = 1135$ s.



(a) y_C. (b) z_C.

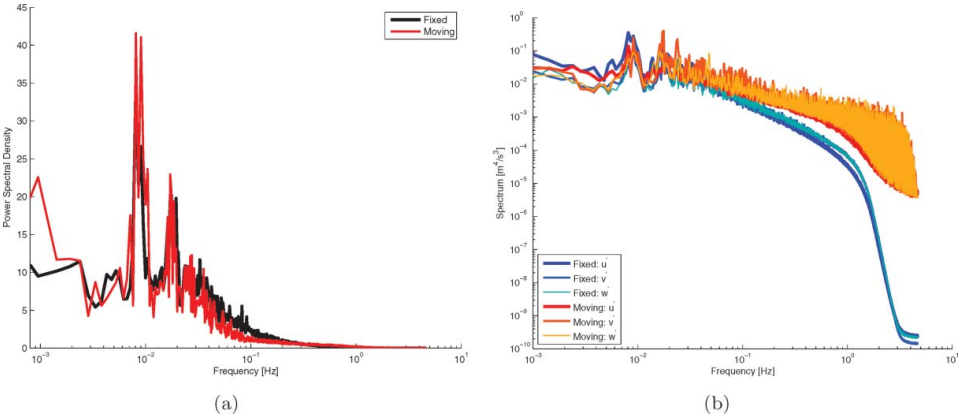
Figure 14. Power spectral density of the wake centre motion with cyclic boundary conditions as calculated using the centre of mass approach and averaged over 7 windows of 35 minutes. Rotational frequency ($1P$) of the wind turbine is marked. (a) Spectrum of y_C. (b) Spectrum of z_C.

no apparent traces of symmetries, axial or polar, which would allow such a transformation to a moving frame of reference. Rather, the wake deficit constantly changes shape and rotational direction, occasionally almost disappearing with only minor deficit experienced by the following turbine.

Furthermore, it is worth noting that the rotational frequency of the wind turbine ($1P = f_{\mathrm{WT}} = 0.29$ Hz) is no longer present as the spectrum is taken $11R$ downstream, as also reported by Lu and Porté-Agel [10].

3.4. *Proper orthogonal decomposition*

The flow field is decomposed in the interval $t \in [1100s - 3200s]$ corresponding to 35 min, within $\pm 2R$ by applying POD to the velocity slices extracted $1R$ upstream the first turbine. Figure 16 shows the normalised POD eigenvalues and the cumulated eigenvalues, i.e.



(a) (b)

Figure 15. Spatially averaged power spectral density for fixed and moving frame of reference. (a) Normalised power spectral density of TKE. (b) Power spectral density of velocity fluctuations.

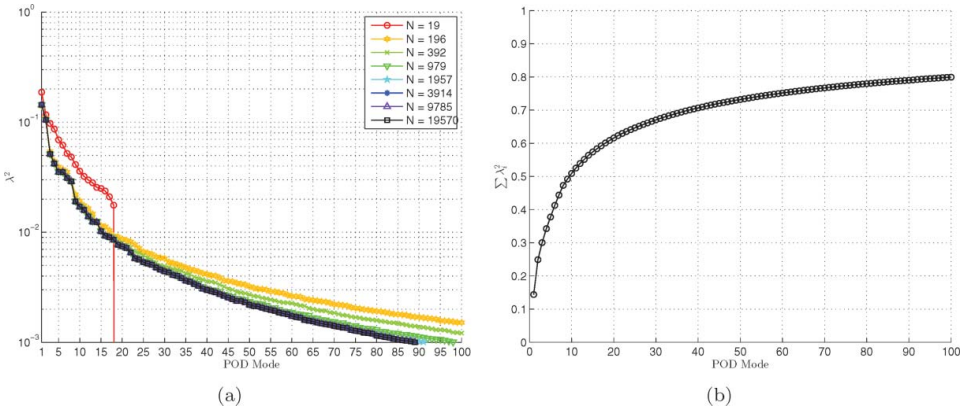
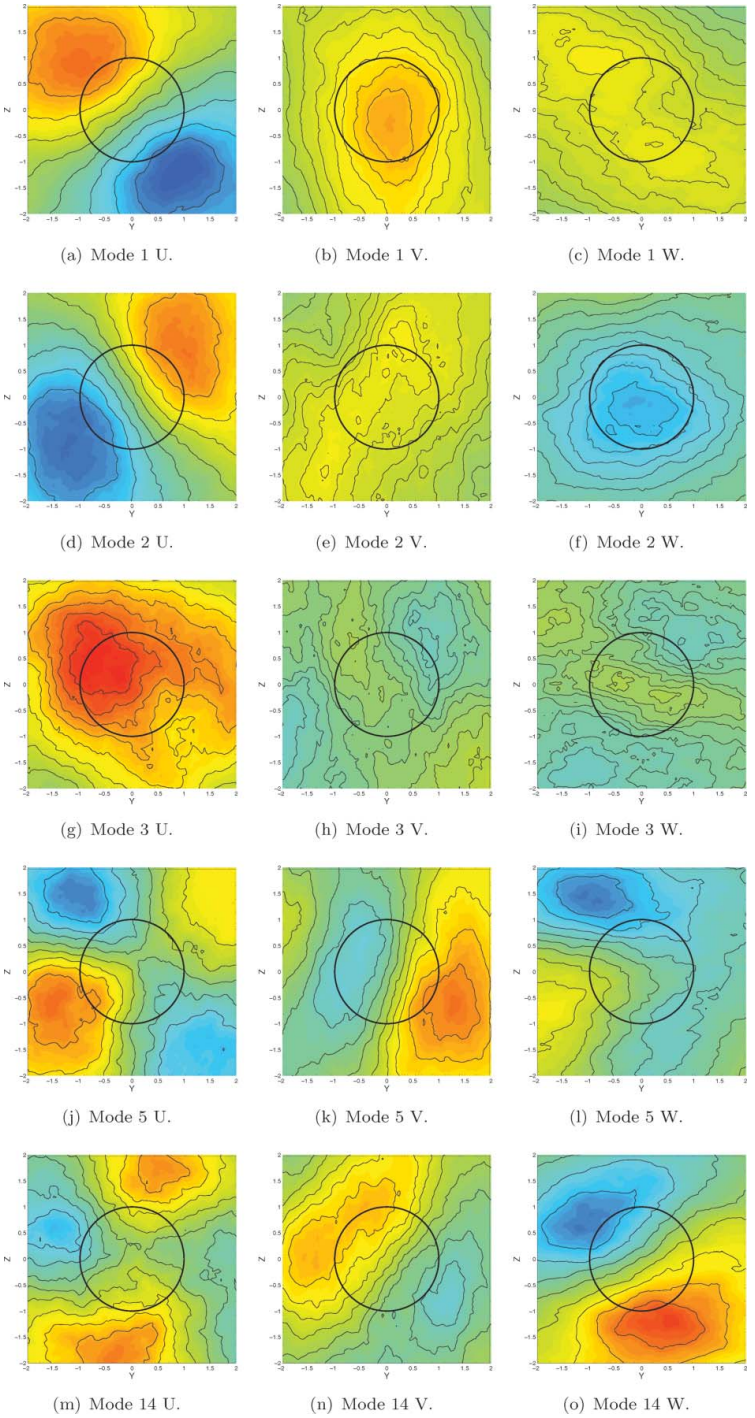


Figure 16. POD eigenvalues. (a) Normalised POD eigenvalues for various numbers of slices in the POD. (c) Cumulated POD eigenvalues for $N = 3914$.

the amount of turbulent kinetic energy captured in the POD modes. The figure on the left includes a convergence study of the normalised POD eigenvalues for various number of included slices within the 35 min. There is no discernible difference in the first 100 POD eigenvalues, when including more than 3914 slices, i.e. using slices with a temporal separation of $\Delta t = 0.5$ s. There are naturally minor differences when looking at the last POD eigenvalues, but even then the difference between using $N = 3914$ and $N = 19,570$ are only $O(10^{-5})$. Using only $N = 979$ yields differences $O(10^{-4})$. This is a clear indication that long time series, and not short discretisations are necessary in order to decompose the flow in an infinite long row of wind turbines. The first three POD modes contain 14.4%, 10.5% and 5.1%, respectively, of the turbulent kinetic energy. The first 10 POD modes hold 51% of the turbulent kinetic energy, while it is necessary to include 403 modes to retain 90% of the total turbulent kinetic energy, corresponding to 0.3% and 10.3% of the 3914 POD modes. The large number of POD needed to reconstruct the flow is a clear indication that the turbulence between the wind turbines have become almost homogeneous.

Figure 17 shows the spatial POD modes 1–5 for all three velocity components. The first two modes of U show a clear dipole structure, but rotated 90° relative to each other due to the orthogonality and mode pairing. Similar mode pairing and dipole structures are evident for POD modes 4 and 5 for V and POD mode 5 for W. POD mode 3 for U, 1 for V and 2 for W portrait a distinct central region. POD modes 4 and 5 for U contain a quadrupole structure, similar to that observed in the production terms involving the streamwise velocity gradient, see Figure 10. The remaining spatial POD modes have less distinct symmetries, but still hold larger regions of turbulent structures. POD modes 12 and 13 for U holds a hexapole structure (not shown). The higher spatial POD modes naturally show smaller and more scattered structures as less organised energy is contained by the modes.

The flow can be reconstructed using a truncated version of Equation (23) with any given number of POD modes, yielding a set amount of reconstructed energy, as given by Figure 16. Figure 18 shows the TKE production terms as well as the total production for a number of reconstructed cases relative to the TKE production of the full flow for the period of 35 min, given in Figure 7. The inplane production terms stemming from the streamwise velocity ($\tau_{xy}\frac{dU}{dy}$ and $\tau_{xz}\frac{dU}{dz}$) as well as those related to the diagonal Reynolds stresses ($\tau_{yy}\frac{dV}{dy}$ and $\tau_{zz}\frac{dW}{dz}$) have a fairly constant convergence towards the full level with minor leaps for



(a) Mode 1 U. (b) Mode 1 V. (c) Mode 1 W.

(d) Mode 2 U. (e) Mode 2 V. (f) Mode 2 W.

(g) Mode 3 U. (h) Mode 3 V. (i) Mode 3 W.

(j) Mode 5 U. (k) Mode 5 V. (l) Mode 5 W.

(m) Mode 14 U. (n) Mode 14 V. (o) Mode 14 W.

Figure 17. Spatial POD modes 1–5 for 35 min for U,V,W.

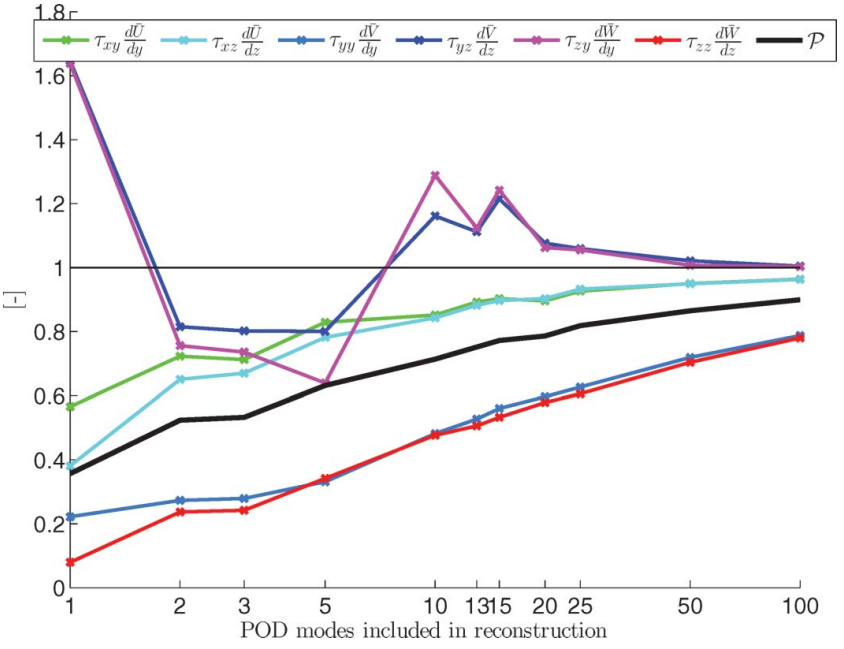


Figure 18. TKE production in the reconstructed flow as function of POD modes used in the reconstruction. The TKE production is given relative to the TKE production for the full flow over 35 min as given in Figure 7. The total TKE production is shown in black. Notice, the logarithmic scale on the x-axis.

the inplane terms. The remaining off-diagonal terms related to the streamwise velocity is less constant as the POD mode pairings clearly counteract each other. POD modes 1 and 2 counteract each other, similar to the modes between 10 and 13, 15 and 20, etc. The black line shows the normalised total TKE production, which also has a steady rise. However, it is worth noting how the first 10 POD modes account for 74% of the total TKE production, but only 51% of the energy variance.

4. Conclusion

The wake interaction and performance of a wind turbine in the interior of a wind farm is investigated using numerical simulations of an idealised case of an infinitely long row of wind turbines by applying cyclic boundary conditions. LES is employed to model the SGS turbulence. The wind turbine is modelled using the AL method coupled with the aeroelastic code Flex5, which calculates loads and performance of a NM80 wind turbine. A controller is implemented, which ensures that the turbine performs as a real wind turbine reacting to the highly complex flow by trying to operate at maximum performance without exceeding the power of the installed generator.

The simulations revealed several distinct peak frequencies $f_1 = 0.009$, $f_2 = 0.017$ and $f_3 = 0.025$, where the first two are related to the spacing of the wind turbines and the latter to the vortex shedding frequency of the turbine as a bluff body with $St = 0.19$. The movement of the wake centre is calculated using a centre of gravity approach. The wake centre motion is highly dynamic and the rotation unpredictable. The simulation proves how the large-scale motions are inherent to the wake interaction deep inside wind farms, and

as such inseparable from the wake dynamics, since the wake constantly morphs between highly complex shapes. Therefore, the wake is not merely a collective entity translated downstream by large scales in the atmospheric turbulence.

Key turbulent quantities have been examined, which showed clear symmetries. These quantities have converged towards a random stationary process, which enables the employment of POD on the data, although long time series are needed. The spatial POD modes reveal both symmetries and mode pairing. The first 10 POD modes include 51% of the turbulent kinetic energy, while 403 modes include 90%. Reconstructing the flow field shows how the first 10 POD modes account for 74% of the total TKE production.

Acknowledgements

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